Introduction to Mol. Sim.

Thursday, 6 January 2022 10:01 AM

Although Statistical mechanics gives us expressions relating inter molecular interactions to thermodynamic properties, it is, in general, not possible to obtain analytical solutions.

i.e. One cannot obtain analytical expressions for the partition functions except for a few idealized cases such as ideal gas, ideal crystal, ising magnet, etc.

Hence we resort to molecular simulations to compute the properties of interest.

Applications of molecular simulations

1) Construction of molecular models.

Propose an empirical model for inter molecular interaction (forcefield)
Compute properties of the material using this model through molecular simulation
Compare your predictions with experimental data
Refine your model parameters to match experimental data.

2) Test thermodynamic theories

Consider the van der Waals equation of state

$$P = \frac{RT}{V-b} - \frac{a}{V^2}$$

$$a = \int_{\sigma}^{\infty} u(r)\, 4\pi r^2\, dr$$

$$b = \frac{2\pi}{3}\sigma^3$$

$\sigma \equiv$ diameter of molecule
$u(r) \equiv$ inter-molecular interaction

Question: Is the vdW EOS a good model for our system?

if the predictions from vdW EOS does not match the experimental data, then there are two possible sources of error.
1) we may have incorrectly estimated the model parameters, σ & $u(r)$
2) The vdW EOS is not a good equation for the thermodynamic state considered

Comparison with experimental data cannot tell us where the error lies.

However, one can perform molecular simulations for the given molecular model & obtain the properties of interest

Molecular simulations give exact results for a given molecular model.

If predictions from the theory (vdW EOS) does not match simulation data, then we can conclusively state that

the theory is inaccurate / not applicable to our situation

Molecular simulations can be considered as a computer experiment or "in silico" experiment

3) Digital microscope

Molecular simulations can provide us with detailed microscopic information, which in general are hard to observe in experiments

4) Discover new phenomena

e.g. discovery of freezing transition in hard-sphere fluids
Alder and Wainwright (1957)

Properties calculated in molecular simulations

1) Thermodynamic properties
e.g. pressure, chemical potential, internal energy, heat capacities, etc.

2) Structural information
e.g radial distribution function, formation of hydrogen bonds etc.

3) Transport properties
e.g. diffusivity, thermal conductivity, viscosity
time correlation functions.

Pros and cons of molecular simulations

Pros
1) obtain detailed microscopic information
2) Usually cheaper to perform compared to physical experiments
3) Easier to investigate extreme conditions e.g. high pressure, low temperature.

Cons
1) Limited to systems that are only nanometer in sizes
2) Limited to phenomena that occur with nano seconds.